A Comparative Analysis of Distance Measures for Ordinal Data in K-Medoids Clustering: Segmenting Guests Based on Service Feature Perceptions in Sri Lanka

A Project Report submitted for the partial fulfilment of the requirements of the Higher National Diploma in Data Science (Part time) Programme

By

Vestika Weerachandra

COHNDDS241P-003

Higher National Diploma in Data Science

National Innovation Centre (NIBM)

Colombo, Sri Lanka

23rd August 2026

Student's Declaration

"I hereby declare that the work presented in this project report was carried out independently by myself and have cited the work of others and given due reference diligently."

Name of the Student: Vestika Weerachandra

Signature:

Date:

Supervisor's Certification

"I certify that the above student carried out his/her project under my supervision and guidance."

Name of the Supervisor: Mr. Ashan Arthanayake

Signature: _____

Date: _____

Dedication

This project is dedicated to everyone who supported me throughout this journey, and to all future researchers who continue to explore data-driven approaches to understanding customer behaviour in Sri Lanka's hospitality industry.

Acknowledgement

I would like to express my sincere gratitude to my supervisor and the lecturers of the Higher National Diploma in Data Science programme at NIBM for their continuous guidance, feedback, and encouragement throughout this project.

I also extend my thanks to House of Travels & Tourism (Pvt) Ltd. for granting access to their customer feedback data, without which this research would not have been possible.

Finally, I am deeply grateful to my family for their patience, encouragement, and unwavering support throughout the duration of this programme.

Executive Summary

This study investigates how customers can be segmented based on their perceptions of hotel service features using structured, ordinal review data collected by House of Travels & Tourism (Pvt) Ltd. This is done using 3,230 reviews which covers 161 hotels and 12 nationalities in Sri Lanka during 2024-2025. The service-rating variables used for this study are ordinal in nature, therefore, this research specifically investigates which distance measure most appropriately represents similarity between customers for clustering purposes, rather than assuming a single distance measure is adequate.

K-Medoids (Partitioning Around Medoids) clustering was applied using three ordinal-appropriate distance measures. The distant measures used were, standard Gower distance, Podani's rank-based ordinal extension of Gower distance, and Walesiak's Generalized Distance Measure (GDM2), which is purpose-built for ordinal item-battery data. Each distance measure was evaluated using Silhouette Score and the Davies-Bouldin Index across a range of cluster counts.

GDM2 produced the best-performing clustering with Silhouette Score = 0.6352. It identified two customer segments, a Highly Satisfied segment with 60.9% of reviews and a Moderately Satisfied segment with 39.1% of reviews. A Kruskal-Wallis test confirmed statistically significant differences between the two segments across all six service features and Overall Rating. The largest effect sizes observed were Overall Rating and Restaurant Service. The findings show that the choice of distance measure meaningfully affects both clustering quality and the resulting segment boundaries.

Table of Contents

Student's Declaration ii
Supervisor's Certification ii
Dedication iii
Acknowledgement iv
Executive Summary v
Table of Contents vi
List of Figures vii
List of Tables viii
Chapter 1: Introduction 1
1.1 Background 1
1.2 Research Problem 3
1.3 Research Questions 3
1.4 Objectives of the Project 4
1.5 Scope of the Research 4
1.6 Justification of the Research 5
1.7 Limitations 6
Chapter 2: Literature Review 7
2.1 Introduction to the research theme 7
2.2 Theoretical explanation about the Key Words in the Topic 7
2.3 Findings by other researchers 9
2.3.1 Data Mining Approaches in Tourism and Hospitality 9
2.3.2 Clustering Approaches for Customer Segmentation 10
2.3.3 Ordinal-Appropriate Distance Measures for Clustering 11
2.4 Research gap 12
2.5 Table for Variables, their definitions and sources 13
2.6 Chapter Conclusion 14
Chapter 3: Methodology 15
3.1 Introduction 15
3.2 Population, sample and Sampling technique 15
3.3 Type of data to be collected and data sources 16
3.4 Data collection tools and plan 16

3.5 Conceptual Framework 17
3.6 Hypothesis 18
3.7 Operationalization Table 18
3.8 Methods of data analysis 19
Chapter 4: Data Analysis 21
4.1 Data Analysis 21
4.1.1 Data Preparation and Pre-Processing 21
4.1.2 Exploratory Data Analysis 22
4.1.3 Distance Measure Computation and Comparison 25
4.1.4 Customer Segment Profiles (GDM2-Based Clustering) 27
4.2 Findings and Interpretation 31
Chapter 5: Discussion and Recommendations 33
5.1 Discussion 33
5.2 Recommendations 34
5.3 Conclusions 35
List of References 36
Appendix 39

List of Figures

Figure 1: Trend of monthly forecast, 2025 (SLTDA 2025a, p.7) 1

Figure 2: Main problem approaches addressed by ML in tourism (Jose, Juan & Rafael 2024) 2

Figure 3: Conceptual Framework 17

Figure 4: Top 10 Nationalities 22

Figure 5: Median Rating of Hotel Service Features 23

Figure 6: Distribution of ratings per Hotel Service Feature 24

Figure 7: Spearman Correlation Between Service Features 25

Figure 8: Silhouette Score Comparison Across Distance Measures 26

Figure 9: Overall Rating Distribution by Cluster 29

Figure 10: Top-Box Percentage by Segment 30

Figure 11: Kruskal-Wallis Effect Size by Service Feature, Ranked 31

List of Tables

Table 1: Table for Variables, their definitions and sources 13

Table 2: Operationalization Table 18

Table 3: Ordinal Encoding 21

Table 4: Dataset Summary 22

Table 5: Central Tendency Summary of Service Features 23

Table 6: Distance Measure Comparison (Silhouette Score and Davies-Bouldin Index) 26

Table 7: Segment Sizes (GDM2-based K-Medoids) 27

Table 8: Segment Sizes (Podani and Gower-based K-Medoids) 27

Table 9: Median and Mode Rating per Feature by Segment 28

Table 10: Overall Rating Summary by Segment 28

Table 11: Top-Box Percentages by Segment (ranked by gap) 29

Table 12: Botton-Box Percentages by Segment (ranked by gap) 30

Table 13: Kruskal-Wallis Test Results and Effect Sizes (Epsilon-Squared), ranked 31

Chapter 1: Introduction

1.1 Background

Tourism industry in Sri Lanka has been a major contribution to the economy, providing significant contribution in GDP, foreign exchange earnings, and employment. In recent years, the tourism industry has shown a significant growth despite challenges such as COVID-19 pandemic, and economic fluctuations. In the year 2024, global tourism had recovered to almost pre-pandemic times (SLTDA 2025a, p.3), and Sri Lanka welcomed 1.17 million tourists in the first half of 2025, a 15.6% increase compared to the same period in 2024 (SLTDA 2025b, p.5).

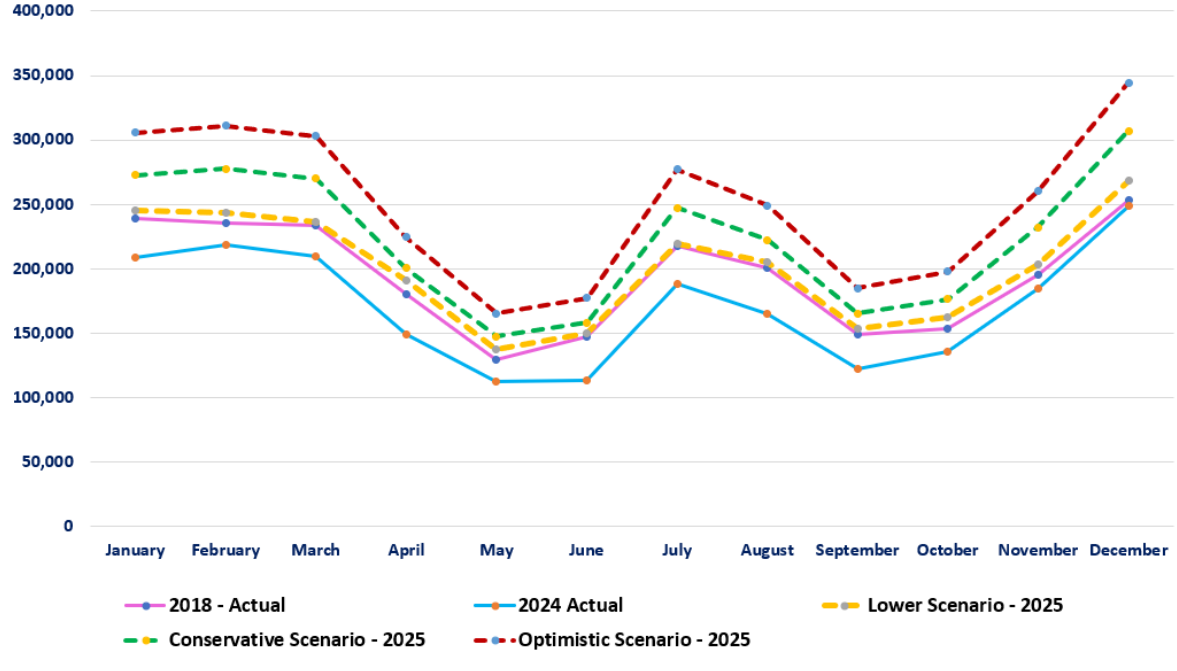


Figure 1: Trend of monthly forecast, 2025 (SLTDA 2025a, p.7)

Customer experience and opinion are crucial for the development of the tourism industry. Globally, the industry has already adapted to various technologies like booking websites. Since tourists utilise websites and social media to share personal opinions or comments on a specific place or service, customer reviews have become a significant factor when deciding places to visit (Karlo & Marina 2022).

Machine Learning (ML) has changed the tourism sector by allowing to analyse large volumes of customer data that allows to recognize trends, understand preferences, and help make data-driven decisions.

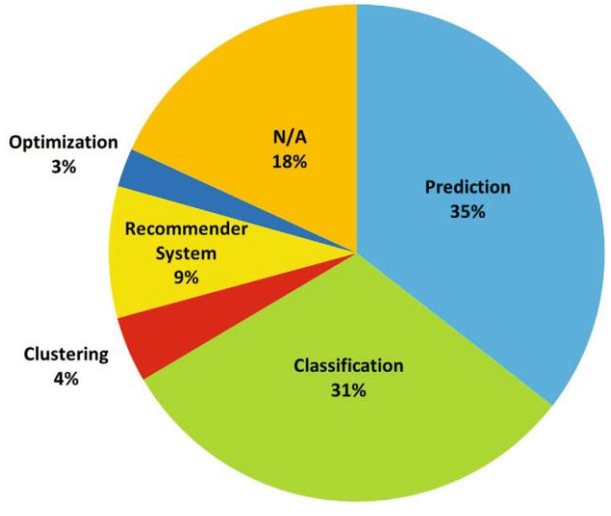


Figure 2: Main problem approaches addressed by ML in tourism (Jose, Juan & Rafael 2024)

Although there are previous studies which usus data mining in hospitality industry, limited work has been done to understand customer behaviour patterns through structured, ordinal hotel services data in Sri Lankan hospitality industry. Further, because such reviews are basically ordinal in nature (e.g., Unacceptable to Excellent) , the choice of the distance measure used within the clustering algorithm has a direct impact on interpreting the customer segments. This is an aspect that is under-explored in the existing tourism analytics literature.

1.2 Research Problem

Understanding customer satisfaction and service expectations is important in improving hotel services. However, there is only limited number of studies about analysis of the customers' behavioural patterns based on structured, ordinal hotel services evaluation and fewer that examine how the choice of distance measure affects the quality of the resulting customer segments. The standard clustering techniques often default to distance measures that assume interval-scaled data, which is not an appropriate assumption when considering ordinal service ratings. Therefore, the objective of this research is to define customer segments using K-Medoids clustering while comparing multiple ordinal-appropriate distance measures, to identify the distance measure that produces the most valid and interpretable customer segments for this type of data.

1.3 Research Questions

1. What are different patterns of customer behaviour identified through hotel service reviews in Sri Lankan hotels?
2. How can the customers be segmented in terms of their perception of hotel service features using K-Medoids (PAM) clustering?
3. Which of the three ordinal-appropriate distance measures out of standard Gower distance, Podani's ordinal extension of Gower distance, or Walesiak's GDM2 - produces the most valid and interpretable customer segments for this dataset, based on internal cluster validity measures?
4. Are the identified customer segments significantly different in how they perceive the individual hotel service features?

1.4 Objectives of the Project

- To identify and analyse customer segments based on hotel service perceptions by applying K-Medoids clustering with multiple ordinal-appropriate distance measures
    - To apply K-Medoids clustering using standard Gower distance, Podani's ordinal extension of Gower distance, and Walesiak's GDM2 distance to group tourists based on their service perceptions.
    - To compare the three distance measures using internal cluster validity measures (Silhouette Coefficient, Davies–Bouldin Index) and determine the most suitable distance measure for this dataset
    - To apply Kruskal-Wallis test to identify differences in the service features rated among the different customer segments identified using the best-performing distance measure.

1.5 Scope of the Research

The research is based on the customer reviews for the hotels collected from the clients of House of Travel & Tourism Pvt Ltd, who had visited multiple properties during their tour in Sri Lanka between 2024 and 2025. Each record represents feedback for a hotel based on Welcome / Check in, Standard of the room, Cleanliness, Restaurant Service, Quality Meals, Attentiveness / Friendliness of Hotel staffs. The dataset also includes customer nationality and number of guests.

The research uses K-Medoids (PAM), with three ordinal-appropriate distance measures (Gower, Podani, GDM2) to identify customer segments and evaluates these using internal cluster validity measures and the Kruskal-Wallis test. The study excludes external review

platforms and focuses only comparative analysis and statistical testing based on the data collected internally by H.O.T (Pvt) Ltd.

1.6 Justification of the Research

The tourism industry, both globally and in Sri Lanka, has been continuously growing in recent years. With many online customer reviews, businesses can utilise this data to understand their customer preferences better and provide more personalised service. In today's technology-driven world, analysing customer behaviour can provide valuable insights that would help to optimise business operations and improve service quality..

In addition to segmenting the customers, this research also makes a contribution to the methodology. Ordinal service ratings are often analysed with distance measures and statistics designed for interval or nominal data, which do not correctly reflect the ordered nature of service ratings data. This study explores the distance measure selection for similar ordinal hospitality datasets which is an under explored area in Sri Lankan hospitality context, by comparing standard Gower distance, Podani's rank-based ordinal extension, and Walesiak's GDM2 (a measure purpose-built for ordinal item-battery data such as multi-attribute service ratings),

1.7 Limitations

- Customer reviews may contain biasness since not all tourists rate their experience consistently.
- The data does not represent the entire tourist population, since it was collected from clients of a single destination management company. Therefore, there may be missing or imbalanced data for certain nationalities.
- The Davies-Bouldin Index was used as a secondary validity check. It assumes Euclidean, centroid-based clusters. Since none of the three distance measures used produce a true Euclidean centroid, this index was approximated on the raw ordinal feature space and should be interpreted alongside the Silhouette Score rather than in isolation.
- The GDM2 distance matrix was computed in R (using the clusterSim package) and imported into the Python analysis environment. This introduced an additional data-handling step compared to the other two distance measures, which were computed natively in Python.
- Some hotels may have limited number of reviews, which could affect the preciseness or results for those properties.
- External factors such as weather, tour guides or duration of stay were not considered in this study.
- The study depends on the accuracy and completeness of the data obtained from the organization's information system.

Chapter 2: Literature Review

2.1 Introduction to the research theme

The digital era has also shown an impact in the tourism industry such that the online reviews of customer's experiences on their travel and stay has become abundant. These reviews are valuable data to understand tourist behaviour and improving the service quality by the organization based on that. With the developments in data analytics and machine learning, researchers have been using these data to derive insights on customer preferences and satisfaction trends. However, limited research has focused on how does the choice of distance/similarity measure affects the quality of customer segments obtained from structured ordinal hotel service review data. This chapter reviews literature relevant to data mining in tourism, clustering approaches for customer segmentation, and ordinal-appropriate distance measures for clustering.

2.2 Theoretical explanation about the Key Words in the Topic

- **Customer Satisfaction**- the extent to which customer expectations are met, not met, or exceeded by the service provided.
- **Customer Behaviour Patterns -** how groups of customers tend to experience and respond to services.
- **Hotel Service Features-** the features of a hotel that influences the customer experience such as cleanliness, room standards, quality of meals and friendliness of the staff.

- **Ordinal Categorical Data** - data that have a meaningful order or rank like the ratings of the hotel from excellent to poor but without a guaranteed equal spacing between the categories.
- **K-Medoids** (Partitioning Around Medoids, PAM) - an unsupervised clustering algorithm where clusters are created around real data points (medoids) rather than computed centroids, and can operate on any precomputed distance matrix.
- **Gower Distance-** a distance measure that calculates a range-normalised dissimilarity per variable and averages across variables, applicable to mixed and ordinal data.
- **Podani's Ordinal Extension of Gower Distance** - a modification of Gower's coefficient that first converts ordinal variables to ranks (with tie correction) before computing the range-normalised distance, so that only rank order, not numeric magnitude, affects the result.
- **GDM2 (Generalized Distance Measure)** - a distance measure developed by Walesiak specifically for ordinal-scale data, based on a generalised correlation coefficient combining Pearson and Kendall correlation.
- **Silhouette Coefficient-** an internal cluster validity measure that checks how similar the observations within the cluster and how clusters differ from each other.
- **Davies–Bouldin Index**- an internal cluster validity measure that checks within-cluster compactness to between-cluster separation, assuming Euclidean centroids.
- **Kruskal-Wallis Test –** a non-parametric, rank-based statistical test used to compare an ordinal or continuous variable across two or more independent groups.

2.3 Findings by other researchers

2.3.1 Data Mining Approaches in Tourism and Hospitality

M. Gunawardana (2019) did a study using Aspect-based Sentimental Analysis to create a summary based on the reviews collected for each specific aspect for Sri Lankan hotels. This aimed to address the difficulty in analysing the large volume of unstructured online review comments for hotels.

Vargas-Calderón et al. (2021) presented a model for the analysis of hospitality service quality through customer reviews with the help of natural language processing and machine learning tools. The researchers used customer reviews of hotels from Booking.com in two cities, namely Bogotá and Madrid. Latent Dirichlet allocation and FastText were used to identify the relevant dimensions of service quality from customer opinions. The results revealed that customer reviews could be analyzed effectively to obtain service quality dimensions.

Kalnaovakul and Promsivapallop (2023) used 25,267 TripAdvisor hotel reviews from 56 hotels in Phuket, Thailand to identify the service quality dimensions. For this purpose, they applied ML and topic modelling through which they identified six major service quality dimensions that showed that different customer groups have different expectations on hotel service quality.

2.3.2 Clustering Approaches for Customer Segmentation

Van Leeuwen and Koole (2022) used Hierarchical Agglomerative Clustering with Gower Distance and Ward's minimum variance linkage. The model identified eight distinct segments based on their behavioural characteristics.

Rezki, Mansouri and Oucheikh (2025) proposed a framework to analyse customer satisfaction by combining Agglomerative Hierarchical Clustering and Random Forest Classifier. The study used the clustering technique to segment customers based on their responses and then used the clustered data to develop a predictive model. The findings showed that customer segmentation identified the distinct satisfaction and dissatisfaction patterns supporting targeted service strategies.

Ziwei Shu et al. (2022), introduced a new approach for segmentation of hotels using the Ordered Weighted Averaging operator, the concept of 2-tuple linguistic representation and K-means clustering. The authors applied the model to the 50 million reviews obtained from TripAdvisor, which covered eight major aspects of hotels. The study showed that clustering techniques can identify customer preference patterns effectively from linguistic hotel evaluations.

2.3.3 Ordinal-Appropriate Distance Measures for Clustering

Szepannek, Aschenbruck and Wilhelm (2025) proposed an extension of the k-prototypes algorithm to Gower distances for mixed-type data containing ordinal variables, demonstrating in a simulation study that this approach achieves clustering quality similar to combining Gower distance with PAM, while offering substantially better runtime performance on large datasets. This directly informed the present study's use of Gower-distance-based K-Medoids as a comparison baseline

Podani (1999) proposed extending Gower's general coefficient of similarity to ordinal characters by first converting ordinal variables to ranks. When some observations have the same values, their average rank is used. The ranked values are then used to calculate the distance using the range-normalised distance formula.

Rodriguez et al. (2021) applied Podani distance to calculate the distance between the patients and the PAM algorithm to perform the clustering. The optimal clusters were determined after using silhouette and the PAM objective for different number of clusters. This provides a relevant background for the present study that uses Podani distance with K-Medoids similarly to this study.

Roszkowska et al. (2024) analysed TripAdvisor airline reviews using GDM2 for multidimensional ordinal customer review data to create an aggregate satisfaction measure. Although the study focused on airline revies, it uses GDM2 with multiple ordinal satisfaction dimensions which has similarity to the methodology used in our study.

Ernst et al. (2025) in a paper directly relevant to this study's design, investigated suitable clustering methods for multivariate ordinal survey data respecting the ordinal nature of the

scale. They compared different treatments of ordinal scale such as nominalisation (ignoring order), standard Gower distance, and Podani's ordinal extension of Gower distance. Their accompanying software also implements GDM2 as an ordinal specific distance, which was adapted to use within K-centroids-style clustering. This paper closely mirrors the present study's comparison of Gower, Podani and GDM2 distances within K-Medoids framework, providing strong methodological support for the research design adopted here.

The previous studies on clustering ordinal data has examined several approaches to measure dissimilarity, including Gower, Podani and GDM2. These measures are different in representing the ordering information in the ordinal data. This methodological difference provides a basis for comparing the clustering performance of the three distance measures in the present study.

2.4 Research gap

Previous studies have successfully applied data mining techniques in analysing hotel review, services and customer satisfaction. But most of the researches have concentrated on analysing unstructured reviews or has applied clustering techniques using distance measures frequently Euclidean-based that do not clearly reflect the ordinal nature of structured service ratings. Although many researches have examined ordinal-appropriate distance measures, limited research has compared ordinal-appropriate distance measures, specifically standard Gower distance, Podani's ordinal extension, and GDM2. within a single real-world hospitality dataset, particularly within the Sri Lankan hospitality context. This study addresses that gap.

2.5 Table for Variables, their definitions and sources

Variable	Definition / Description	Source
Booking No	Unique identification number assigned for each booking	H.O.T (Pvt) Ltd Customer Feedback Database
Hotel ID	Unique code assigned to each hotel in the database	H.O.T (Pvt) Ltd Customer Feedback Database
Hotel Name	Name of the hotel where the guests stayed	H.O.T (Pvt) Ltd Customer Feedback Database
Nationality	Country of origin of the guests	H.O.T (Pvt) Ltd Customer Feedback Database
Welcome / Check-in	Guest rating on the welcome and check-in experience	H.O.T (Pvt) Ltd Customer Feedback Database
Standard of the Room	Guest rating on the comfort, facilities, and overall standard of the room	H.O.T (Pvt) Ltd Customer Feedback Database
Cleanliness	Guest rating on the cleanliness of the hotel and rooms	H.O.T (Pvt) Ltd Customer Feedback Database
Restaurant Service	Guest rating on the quality and efficiency of restaurant services	H.O.T (Pvt) Ltd Customer Feedback Database
Quality of Meals	Guest rating on the quality and taste of food provided	H.O.T (Pvt) Ltd Customer Feedback Database
Attentiveness / Friendliness of Hotel Staff	Guest rating on the hospitality and friendliness of staff members	H.O.T (Pvt) Ltd Customer Feedback Database
Overall Rating	Overall satisfaction rating given by the guest for the hotel stay	H.O.T (Pvt) Ltd Customer Feedback Database

Table 1: Table for Variables, their definitions and sources

2.6 Chapter Conclusion

This chapter reviewed key literature relevant to data mining in tourism and hospitality, clustering approaches for customer segmentation, and ordinal-appropriate distance measures. The reviewed literature demonstrates growing interest in structured, data-driven customer segmentation, but also reveals a clear gap which is the limited comparison of ordinal-appropriate distance measures, specifically Gower, Podani, and GDM2, applied to a single hospitality dataset. This study addresses that gap using customer feedback data from House of Travels & Tourism (Pvt) Ltd.

.

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the methodology used to achieve the research objectives. This explains the processes used in the study for data collection, preparation and analysis. For the purpose of this study, I have used the secondary data extracted from the customer feedback database and booking reservation dataset of House of Travels & Tourism (Pvt) Ltd, which includes hotel review information from tourists of various nationalities. K-Medoids clustering was applied using three ordinal-appropriate distance measures, standard Gower distance, Podani's ordinal extension of Gower distance, and Walesiak's GDM2, and the resulting clustering segments were compared using internal cluster validity measures. The best-performing distance measure was then used to identify customer segments, and the Kruskal-Wallis test was applied to examine differences among segments across service features.

3.2 Population, sample and Sampling technique

The data for this study is collected from the customer feedback database if H.O.T (Pvt) Ltd, which covers hotel bookings from variety of hotels and tourists of different nationalities. The dataset includes 3230 reviews recorded over the year 2024 and 2025 by different nationality for various hotels. Since the study utilises all the available data, it represents a comprehensive collection of customer reviews.

3.3 Type of data to be collected and data sources

The data is collected from a destination management company called House of Travels & Tourism (Pvt) Ltd hence this is a secondary data. This has fields such as hotel details, tourist nationality, number of guests and ratings for different aspects of the hotel stay.

3.4 Data collection tools and plan

Since the data was obtained directly from the internal customer feedback system of H.O.T (Pvt) Ltd, no primary data collection tools such as surveys or interviews were used. The data extraction process involved retrieving all relevant records from the database, linking the hotel ID with the relevant hotel names, and extracting the nationality and the number of guests of each booking from the reservation system of the company.

3.5 Conceptual Framework

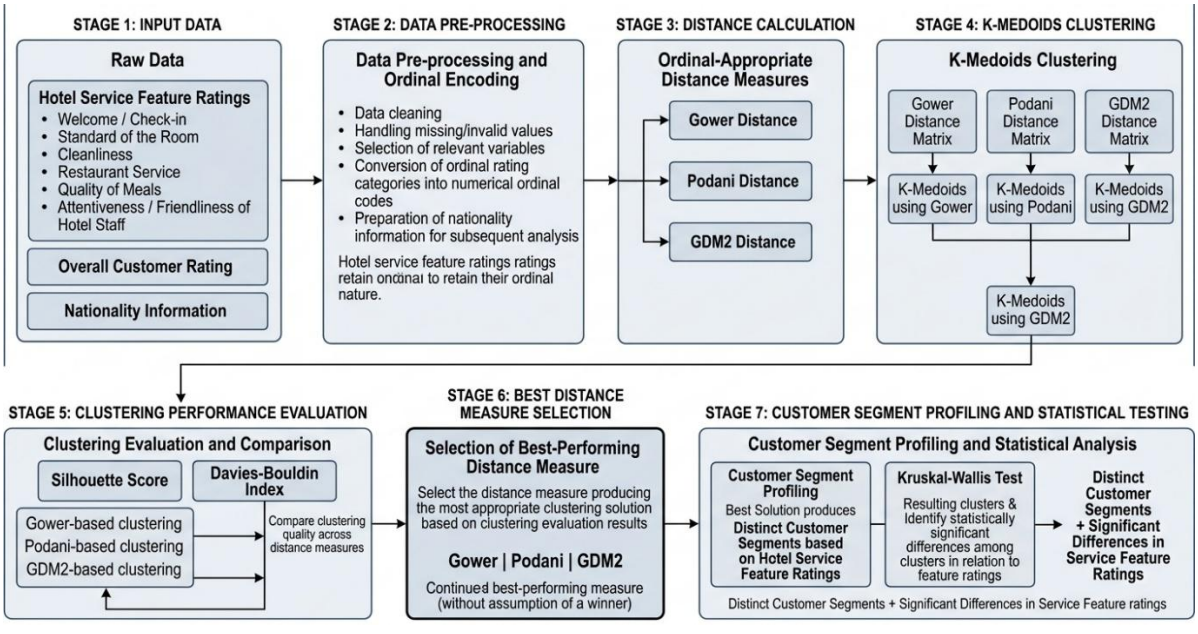


Figure 3: Conceptual Framework

The conceptual framework for this study would be proceeded as follows;

- Raw data are pre-processed and both the service feature ratings and overall customer ratings are ordinally encoded
- Three ordinal-appropriate distance matrices (Gower, Podani, GDM2) are calculated from the encoded service features
- K-Medoids clustering is applied independently using each distance matrix
- The three resulting clustering results are evaluated and compared using Silhouette Score and Davies-Bouldin Index
- The best-performing distance measure is selected
- The resulting customer segments are profiled and tested for statistically significant differences using the Kruskal-Wallis test.

3.6 Hypothesis

H0 (Null Hypothesis): There is no statistically significant difference in hotel service feature ratings between the customer segments identified through K-Medoids clustering.

H1 (Alternative Hypothesis): There is a statistically significant difference in hotel service feature ratings between the customer segments identified through K-Medoids clustering.

3.7 Operationalization Table

Variable	Definition	Indicators	Measures
Welcome/Check-in	Rating of arrival/departure experience	Friendliness, ease of process	Ordinal: 0-4
Standard of Room	Rating of room comfort and condition	Room size, bed comfort, amenities	Ordinal: 0-4
Cleanliness	Rating of hygiene and maintenance	Absence of dirt, hygiene standards	Ordinal: 0-4
Restaurant Service	Rating of dining service quality	Timely delivery, staff attentiveness	Ordinal: 0-4
Quality Meals	Rating of food taste and quality	Taste, freshness, variety	Ordinal: 0-4
Attentiveness/Friendliness	Rating of staff helpfulness	Staff attitude, responsiveness	Ordinal: 0-4
Overall Rating	Overall satisfaction score	General satisfaction level	Ordinal: 0-4

Table 2: Operationalization Table

3.8 Methods of data analysis

The data obtained from the H.O.T (Pvt) Ltd.'s customer feedback system will be cleaned and pre-processed to deal with missing or inconsistent values, including treating placeholder values (e.g., 'select') as missing and imputing them using the mode. Duplicate records were identified using the combination of booking number and hotel ID. Service ratings were then ordinally encoded (Unacceptable=0 through Excellent=4).

Exploratory Data Analysis (EDA) was conducted using measures of central tendency (mean, median, mode) appropriate to ordinal data, along with rating distributions per feature. Spearman's rank correlation was used to examine relationships between service features which only depends on the rank order rather than the numeric magnitude of the ordinal data.

Three ordinal-appropriate distance matrices were then computed on the ordinally-encoded service features: (1) standard Gower distance, computed as the range-normalised absolute difference averaged across features, (2) Podani's ordinal extension of Gower distance, computed identically but on rank-transformed, and (3) Walesiak's GDM2 distance, computed in R using the clusterSim package and imported into the Python analysis environment.

K-Medoids (PAM) clustering was applied independently using each of the three distance matrices, testing cluster counts from k=2 to k=8. For each distance measure and cluster count, the Silhouette Score was computed using the corresponding precomputed distance matrix. The Davies-Bouldin Index was also computed for each solution as a secondary, supporting measure. As none of the three distances yield a native Euclidean centroid, this index was approximated on the raw ordinal feature space uniformly across all three

distance measures, and was treated as secondary to the Silhouette Score in the final decision.

The distance measure producing the highest Silhouette Score was selected as the basis for further analysis. Using the resulting customer segments, profiling was conducted using median and mode ratings and Top-Box/Bottom-Box percentage analysis which is a standard approach in survey research for summarising ordinal/Likert-type response data without assuming interval spacing. Finally, the Kruskal-Wallis test, a non-parametric, rank-based test appropriate for comparing ordinal data across groups, was applied to each service feature to determine whether the identified segments differ significantly, supplemented by the epsilon-squared effect size to assess the practical magnitude of these differences.

Chapter 4: Data Analysis

4.1 Data Analysis

4.1.1 Data Preparation and Pre-Processing

The customer feedback dataset was prepared by merging hotel service ratings with hotel names and customer nationality/booking information. Duplicate records were identified using the combination of booking number and hotel ID. Placeholder values without meaningful rating information were treated as missing and imputed using the mode. Ordinal ratings were then encoded to numerical values as shown in Table 3, enabling quantitative analysis while preserving the ordinal category order.

Rating Category	Numerical Value
Unacceptable	0
Poor	1
Satisfactory	2
Good	3
Excellent	4

Table 3: Ordinal Encoding

4.1.2 Exploratory Data Analysis

Measure	Value
Total Reviews	3230
Number of Hotels	161
Number of Nationalities	12
Number of Service Features	6

Table 4: Dataset Summary

When the distribution of customer nationalities was analysed, it showed that there were customers from multiple nationalities which showed the diversity of dataset. But certain nationalities have contributed to higher number of reviews compared to others. This imbalance was considered during later analysis and is discussed as a limitation.

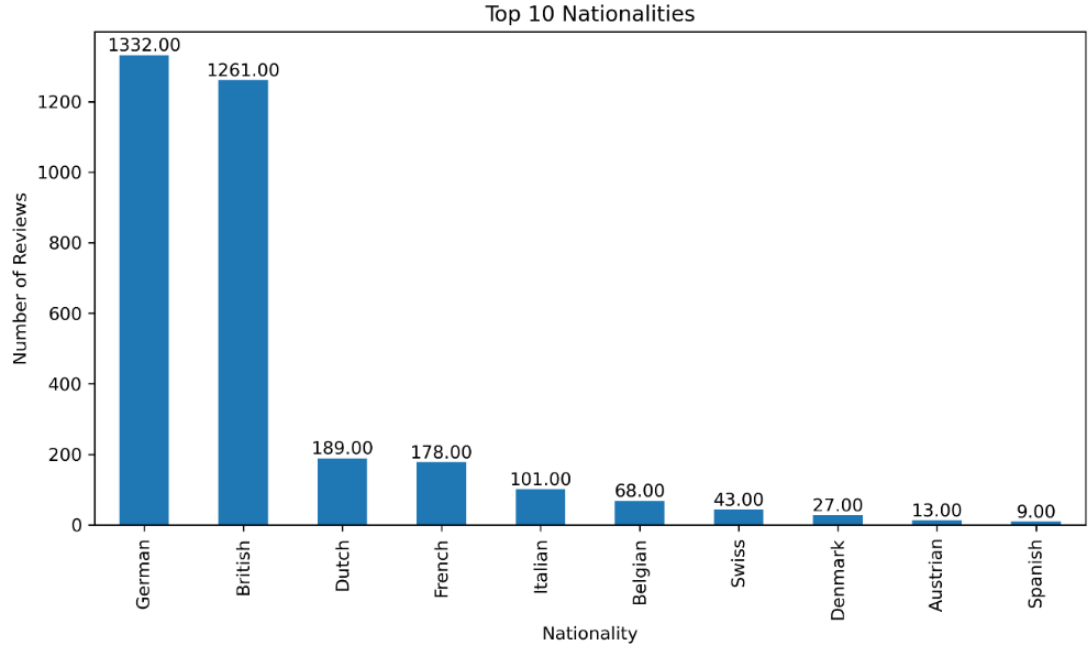


Figure 4: Top 10 Nationalities

Due to the ordinal nature of the rating scale, mean, median, and mode were examined for each service feature (Table 5). While the mean is presented for interpretability and comparability with conventional summary charts, it treats the ordinal scale as if evenly

spaced, which is a simplification. Median and mode values, which do not rely on this assumption, are considered the more statistically appropriate summary measures for this data type.

Attributes	Mean	Median	Mode	Mode (label)
Welcome / Check in	3.64	4	4	Excellent
Standard of the room	3.44	4	4	Excellent
Cleanliness	3.51	4	4	Excellent
Restaurant Service	3.5	4	4	Excellent
Quality Meals	3.41	4	4	Excellent
Attentiveness / Friendliness of Hotel staffs	3.63	4	4	Excellent
Overall Rating	3.45	4	4	Excellent

Table 5: Central Tendency Summary of Service Features

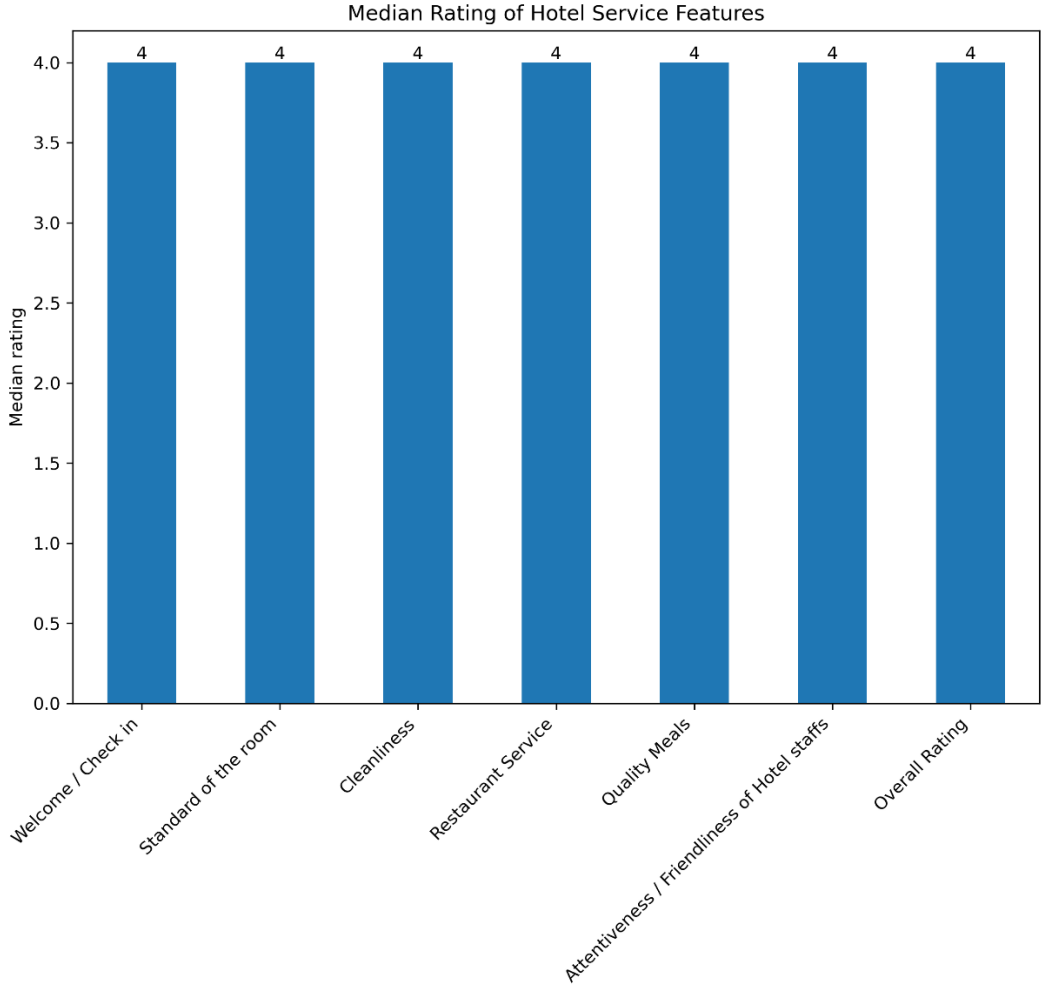


Figure 5: Median Rating of Hotel Service Features

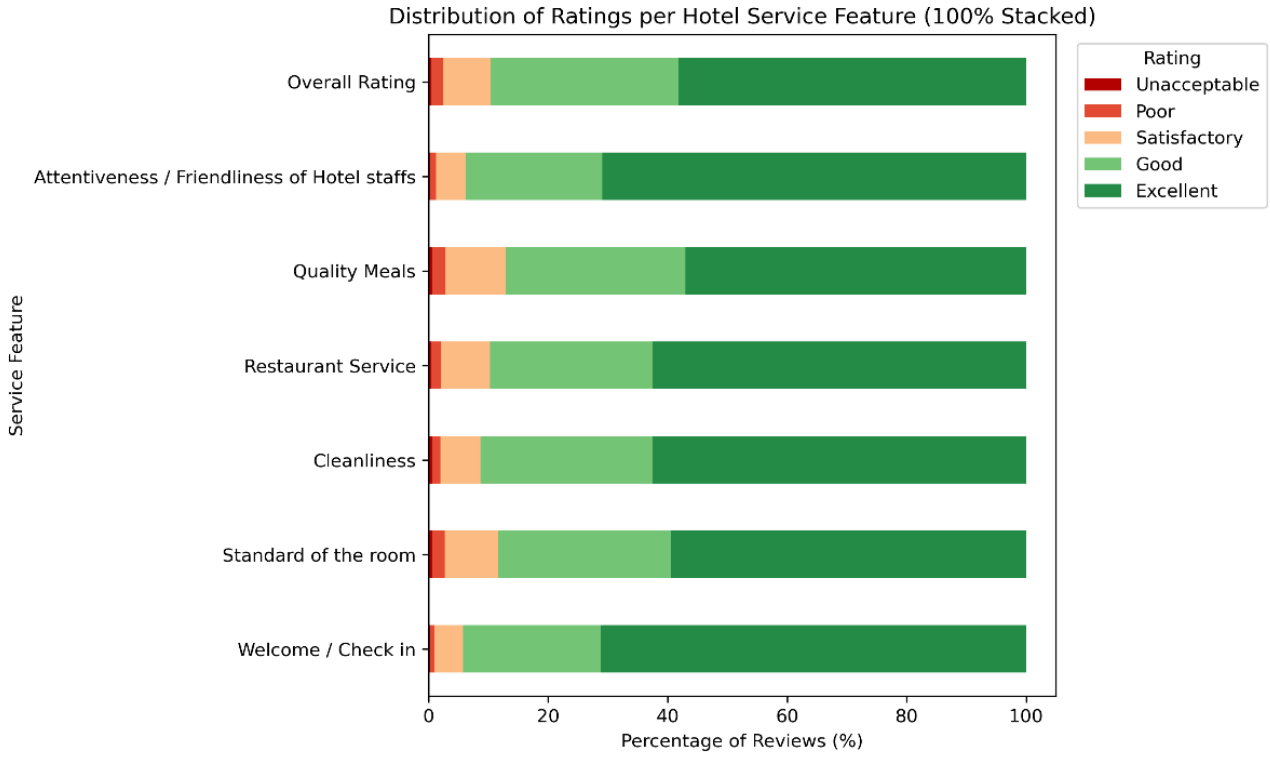


Figure 6: Distribution of ratings per Hotel Service Feature

A Spearman correlation was carried out to find the relationships between the hotel service features. All the variables showed positive relationships. Restaurant service had the strongest correlation with Overall Rating (rho=0.74) and Quality meals (rho=0.73). Standard of the room and Cleanliness had the next strongest correlation with rho=0.72. The strongest relationships with Overall rating were restaurant Service followed by Standard of the room.

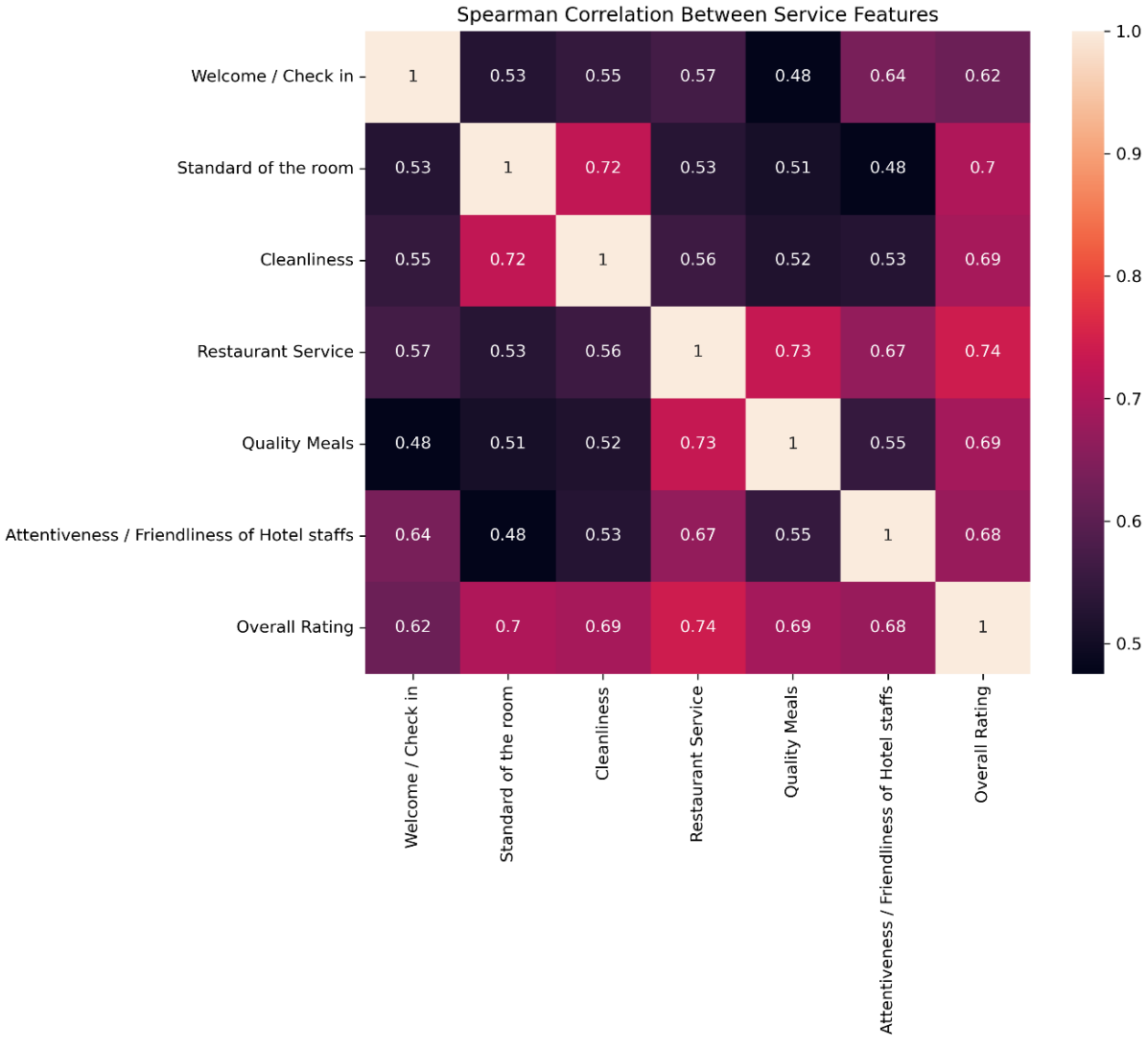


Figure 7: Spearman Correlation Between Service Features

4.1.3 Distance Measure Computation and Comparison

Three ordinal-appropriate distance matrices were computed on the six service features. The distant matrices were standard Gower distance, Podani's rank-based ordinal extension of Gower distance, and Walesiak's GDM2 distance (computed in R and imported for analysis). K-Medoids clustering was applied independently using each distance matrix, testing

cluster counts k=2 to k=8. For all three distance measures, k=2 produced the highest Silhouette Score, showing consistent evidence for two-segment structure in the data.

Distance Measure	Clusters (k)	Silhouette Score	Davies-Bouldin Index (approx.)
Gower	2	0.5991	0.9645
Podani (Ordinal-Gower)	2	0.6321	0.9870
GDM2 (Walesiak)	2	0.6352	0.9654

Table 6: Distance Measure Comparison (Silhouette Score and Davies-Bouldin Index)

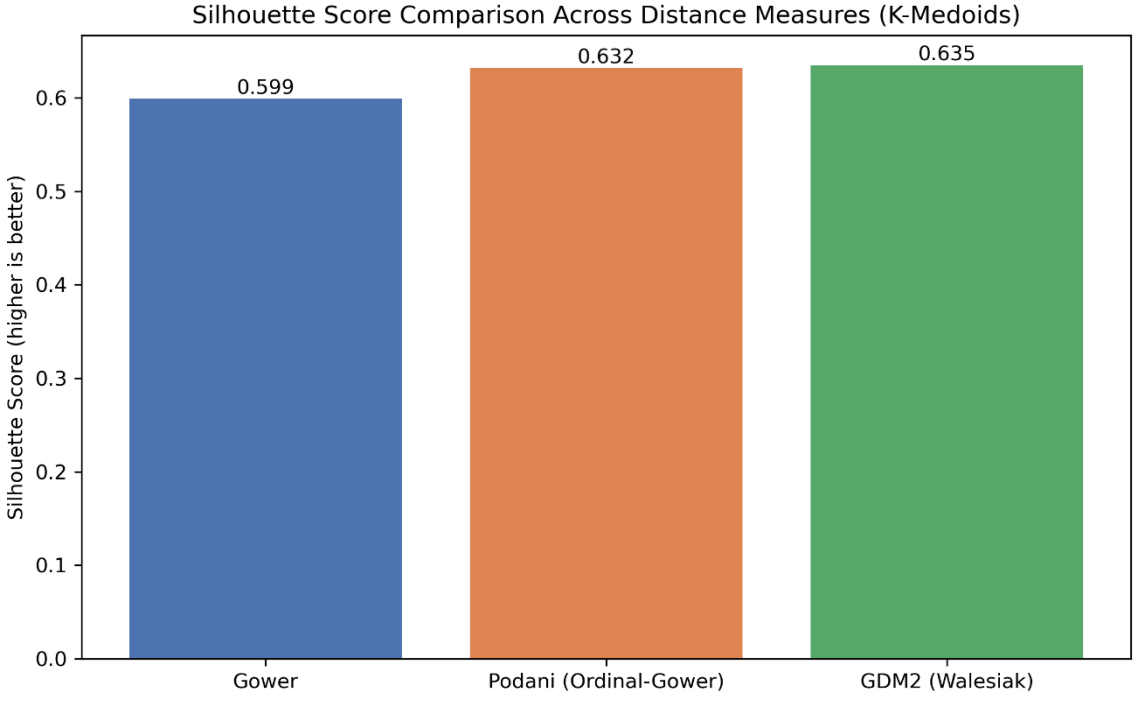


Figure 8: Silhouette Score Comparison Across Distance Measures

Silhouette Score was treated as the primary decision criterion. GDM2 produced the best clustering result with Silhouette = 0.6352, followed by Podani (0.6321) and standard Gower (0.5991). The Davies-Boudlin Index produced a different ranking, with Podani having the least value (0.9870) although it was second best with Silhouette score. This

discrepancy is due to that Podani's clustering was formed using the rank space and DBI was calculated using its raw-code space while reemphasise that Silhouette Score is more reliable metric for this comparison. Therefore, GDM2 was selected as the distance measure and carried forward for segment profiling and hypothesis testing.

4.1.4 Customer Segment Profiles (GDM2-Based Clustering)

Segment	Number of Reviews	Percentage
Segment 0 (Highly Satisfied)	1,966	60.87%
Segment 1 (Moderately Satisfied)	1,264	39.13%

Table 7: Segment Sizes (GDM2-based K-Medoids)

Distance Measure	Cluster 0 (%)	Cluster 1 (%)
Podani	67.59	32.41
Gower	70.09	29.91

Table 8: Segment Sizes (Podani and Gower-based K-Medoids)

Median and mode ratings were identical across all six service features within each segment. Segment 0 showed a median and mode of Excellent (4) for every feature, and Segment 1 showed a median and mode of Good (3) for every feature.

Service Feature	Segment 0 (Median/Mode)	Segment 1 (Median/Mode)
Welcome/Check-in	Excellent	Good
Standard of the Room	Excellent	Good

Service Feature	Segment 0 (Median/Mode)	Segment 1 (Median/Mode)
Cleanliness	Excellent	Good
Restaurant Service	Excellent	Good
Quality Meals	Excellent	Good
Attentiveness/Friendliness	Excellent	Good

Table 9: Median and Mode Rating per Feature by Segment

Overall Rating also showed similar pattern. Segment 0 had a high Overall Rating of median=4 and mean = 3.90, than Segment 1 which had a median = 3 and mean= 2.75. This shows a clear distinction in overall guest satisfaction between the two segments.

Segment	Mean	Median	Count
Segment 0	3.90	4.0	1,966
Segment 1	2.75	3.0	1,264

Table 10: Overall Rating Summary by Segment

Further, the overall rating distribution proved the pattern seen by the mean and median with the overall share of each rating in the clusters..

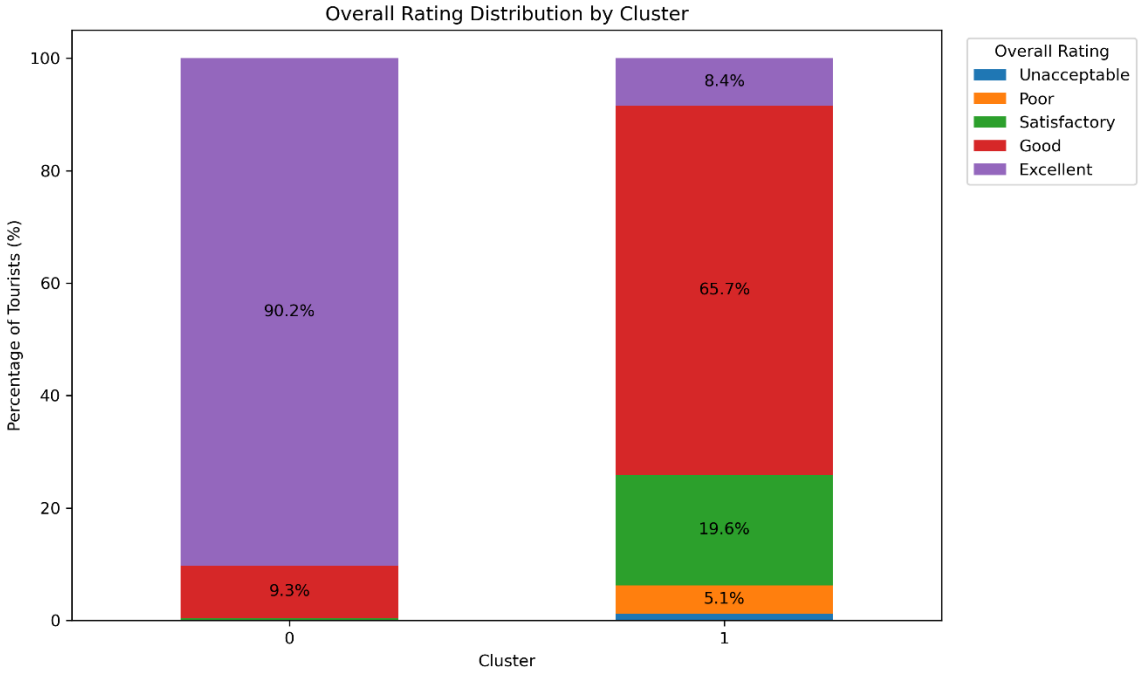


Figure 9: Overall Rating Distribution by Cluster

Further, Top-Box (percentage rating Good or Excellent) and Bottom-Box (percentage rating Unacceptable or Poor) percentages were computed per feature.

Service Feature	**Segment 0 Top-Box %**	**Segment 1 Top-Box %**	**Top-Box Gap**
Quality Meals	98.6	69.2	29.4
Standard of the Room	98.8	72.2	26.6
Restaurant Service	99.6	74.3	25.3
Cleanliness	99.6	78.6	21.0
Attentiveness/Friendliness	99.7	84.5	15.2
Welcome/Check-in	99.4	86.3	13.1

Table 11: Top-Box Percentages by Segment (ranked by gap)

Service Feature	Segment 0 Bottom-Box %	Segment 1 Bottom-Box %	Bottom-Box Gap
Quality Meals	0.1	7.0	6.9
Standard of the Room	0.1	6.9	6.8
Restaurant Service	0.0	5.4	5.4
Cleanliness	0.1	4.8	4.7
Attentiveness/Friendliness	0.1	3.1	3.0
Welcome/Check-in	0.2	2.4	2.2

Table 12: Botton-Box Percentages by Segment (ranked by gap)

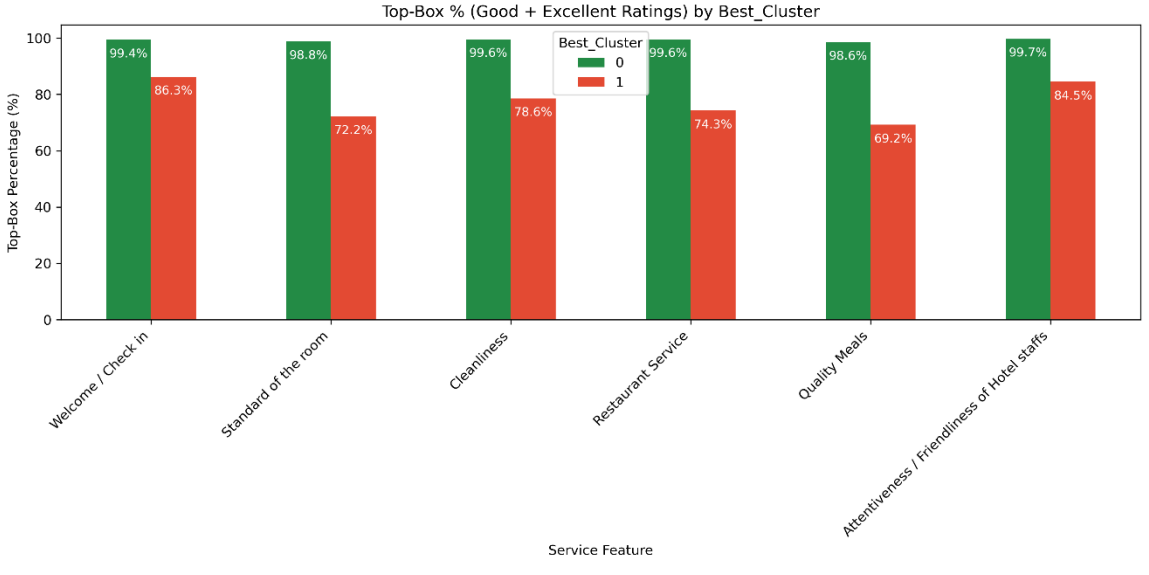


Figure 10: Top-Box Percentage by Segment

Quality Meals and Restaurant Service showed the largest Top-Box gaps between segments *with* 29.4 and 25.3 percentage points respectively. Welcome/Check-in and Attentiveness/Friendliness showed the smallest gaps with 13.1 and 15.2 percentage points. This shows that even Segment 1 (Moderately Satisfied) guests rate check-in experience and staff friendliness relatively well, whereas dining-related experience (meals and restaurant service) is where the clearest separation between segments occurs.

4.2 Findings and Interpretation

Kruskal-Wallis test was conducted for each service feature and Overall Rating to check whether ratings differ significantly between the two GDM2-based segments. The results showed that all seven attributes are statistically significant results ($p<0.001$), which led to reject the null hypothesis (H0) stated in Section 3.6 for every feature tested. Therefore, the alternative hypothesis (H1): a statistically significant difference exists between segments, is accepted.

Service Feature	H-Statistic	p-value	Epsilon-Squared
Overall Rating	2109.16	< 0.001	0.6532
Restaurant Service	1966.91	< 0.001	0.6091
Cleanliness	1721.27	< 0.001	0.5331
Quality Meals	1605.64	< 0.001	0.4973
Standard of the Room	1534.54	< 0.001	0.4752
Attentiveness/Friendliness	1532.13	< 0.001	0.4745
Welcome/Check-in	1416.82	< 0.001	0.4388

Table 13: Kruskal-Wallis Test Results and Effect Sizes (Epsilon-Squared), ranked

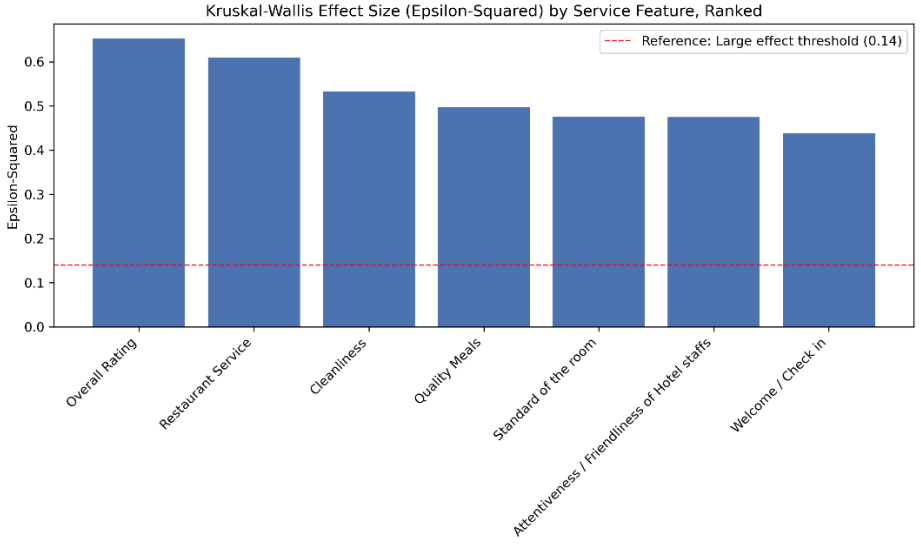


Figure 11: Kruskal-Wallis Effect Size by Service Feature, Ranked

Considering the large sample size (n=3,230), statistical significance was expected for all features. Therefore, the epsilon-squared was conducted to check the magnitude of each difference. The effect size for all cases were more than 0.14, which is the conventional threshold for ‘large’ effect size. This proves that there is a substantial difference between the segments rather than just statistical difference when all features are considered. Overall Rating and Restaurant Service has the largest effect sizes with 0.6532 and 0.6091 respectively while Welcome/Check-in has the lowest effect size though its large. This outcome also aligns with what was represented through Top-Box gap analysis in Section 4.1.4.

Regarding the research questions raised in Chapter 1:

1. Different behaviours of customers were observed through the hotel service reviews.
2. Customers were segmented successfully using K-Medoids into two segments
3. Among the three distance measures, GDM2 was chosen as the most appropriate one according to the Silhouette Score
4. There were significant different between the segments that were found for all of the service features.

Chapter 5: Discussion and Recommendations

5.1 Discussion

The results obtained in this study confirmed theoretical results presented in the literature reviewed in Chapter 2 where it is stated that GDM2 is better than both standard Gower distance and the ordinal extension of Gower distance by Podani. GDM2 was specifically designed by Walesiak (1999) for handling ordinal-scale, item-battery data, such as the six uniformly-scaled service ratings used in this study, while Gower and Podani are more general treatments of ordinal data. Gower and Podani considers the order of the ordinal ratings but GDM2 measure goes one step further and considers the relations of the rating variables as well. This is reflected in the Silhouette Score result as well since the score improves from Gower (0.5991), to Podani (0.6321), and reaches its peak value in GDM2 (0.6352). Therefore, it can be concluded that the GDM2 measure produced the best-defined clusters among the three measures.

All the three distant measures produced two segments but the number of customers in each segments differed. Gower produced segments of 70.09% and 29.91%, Podani produced segments of 67.59% and 32.41% and GDM2 produced segments of 61% and 39%. This shows that although the number of clusters are uniform the segmentation of the customers may vary based on the distant measure used.

The findings at the feature level shows that Quality Meals and Restaurant Service differentiate the two segments while Welcome/Check-in and Staff Attentiveness show comparatively small gaps. This aligns well with the study done by Kalnaovakul and Promsivapallop (2022) that the highly satisfied and moderately satisfied customers have

different expectations on each of the service quality dimensions. And in this dataset, the dining experience is the primary driver separating the highly satisfied from moderately satisfied.

One of the limitations acknowledged throughout the study was the Davies-Bouldin Index, because none of the three distance measures produce a native Euclidean centroid, this index could only be approximated on the raw feature space. This introduced a degree of inconsistency, specifically for Podani distance since it uses the rank space but the DBI is calculated in raw-code space. This could be the reason for the relatively low DBI value of Podani compared to its relatively high Silhouette Score. Due this factor, Silhouette Score was considered as the primary and DBI was considered as secondary supporting metric in comparing ordinal clusters of this type

5.2 Recommendations

- House of Travels & Tourism (Pvt) Ltd, and partner hotels, should prioritise improvements to Quality Meals and Restaurant Service, as these features showed the largest gap between the Highly Satisfied and Moderately Satisfied segments, indicating the greatest potential impact on overall guest satisfaction.
- For future ordinal clustering studies similar to structured hospitality review data, GDM2 is recommended as the preferred distance measure due its design for ordinal item-battery data and its superiority in this study.
- Future research should investigate the association between customer segments and nationality (not explored in this study) to make further service improvement strategy recommendations by nationality of guests.

- Future research might also further compare the model-based clustering methods (Latent Class Analysis) that were mentioned in the literature review and are a fundamentally different (distance-free) paradigm for clustering ordinal data.

5.3 Conclusions

This study was conducted to identify which ordinal appropriate distant measure out of standard Gower distance, Podani's ordinal extension and GDM2, produces the most valid and interpretable customer segments using K-Medoids clustering. This was applied to the structured hotel service review data from House of Travels & Tourism (Pvt) Ltd. GDM2 was identified as the best-performing distance measure based on the Silhouette Score, which showed two segment structure of Highly Satisfied (60.9%) and Moderately Satisfied (39.1%) guests. The Kruskal-Wallis test confirmed that these segments differ significantly, with large effect sizes, across all six service features and Overall Rating. Quality Meals and Restaurant Service identified as the features most strongly differentiating the two segments.

These results provide House of Travels & Tourism (Pvt) Ltd an empirically grounded basis for focusing the service improvement initiatives and offer researchers and practitioners using clustering analysis with ordinal hospitality data a methodological guidance. This analysis should be extended in the future to look into the association between customer segments and nationality. Also, this could explore model-based clustering alternatives such as Latent Class Analysis.

List of References

Alejandro, R., et al. (2021) ’ Deploying unsupervised clustering analysis to derive clinical phenotypes and risk factors associated with mortality risk in 2022 critically ill patients with COVID-19 in Spain’. Available at: https://doi.org/10.1186/s13054-021-03487-8 (Accessed: 22 August 2026)

Ernst, D., Ortega Menjivar, L., Scharl, T. and Grun, B. (2025) 'Ordinal Clustering with the flex-Scheme', Austrian Journal of Statistics, 54(3), pp. 100-123. Available at: https://doi.org/10.17713/ajs.v54i3.2083 (Accessed: 22 August 2026).

Ewa, R., Bartłomiej, J., and Marta, K., 2024,’ A SYNTHETIC MEASURE FOR RANKING OVERALL SATISFACTION BASED ON MULTIDIMENSIONAL ORDINAL DATA. THE CASE OF AIRLINES ON THE TRIPADVISOR WEBSITE’. Available at: 10.37190/ord240312. (Accessed: 22 August 2026,)

Gunawardana, M. (2019) Aspect based sentimental analysis of Sri Lankan Hotels. Available at: https://dl.ucsc.cmb.ac.lk/jspui/handle/123456789/4367 (Accessed: 11 November 2025).

Kalnaovakul, K. and Promsivapallop, P. (2022) 'Hotel service quality dimensions and attributes: An analysis of online hotel customer reviews', Tourism and Hospitality Research. Available at: https://doi.org/10.1177/14673584221145819 (Accessed: 25 July 2026).

Karlo Puh and Marina Bagic Babac (2022) ‘Predicting sentiment and rating of tourist reviews using machine learning’. Available at: https://doi.org/10.1108/JHTI-02-2022-0078 (Accessed: 05 November 2025)

Podani, J. (1999) 'Extending Gower's General Coefficient of Similarity to Ordinal Characters', Taxon, 48(2), pp. 331-340.

Rezki, N., Mansouri, M. and Oucheikh, R. (2025) 'Deciphering Customer Satisfaction: A Machine Learning-Oriented Method Using Agglomerative Clustering for Predictive Modeling and Feature Selection', Management Systems in Production Engineering. Available at: https://doi.org/10.2478/mspe-2025-0007 (Accessed: 29 July 2026).

Rik van Leeuwen and Ger Koole (2022), 'Data-driven market segmentation in hospitality using unsupervised machine learning'. Available at: https://doi.org/10.1016/j.mlwa.2022.100414 (Accessed: 25 July 2026)

Shu, Z., Carrasco Gonzalez, R.A., Portela Garcia-Miguel, J. and Sanchez-Montanes, M. (2023) 'Clustering using ordered weighted averaging operator and 2-tuple linguistic model for hotel segmentation: The case of TripAdvisor', Expert Systems with Applications. Available at: https://doi.org/10.1016/j.eswa.2022.118922 (Accessed: 25 July 2026).

Sri Lanka Tourism Development Authority (SLTDA) (2025) Growth Scenarios for Tourism to Sri Lanka 2025. Available at: https://www.sltda.gov.lk (Accessed: 05 November 2025).

Sri Lanka Tourism Development Authority (SLTDA) 2025b, 'Year in review, January to June 2025;, SLTDA, Available at: https://www.sltda.gov.lk/storage/common_media/Year_In_Review_2025_final_.pdf (Accessed: 05 November 2025)

Szepannek, G., Aschenbruck, R. and Wilhelm, A. (2024) 'Clustering large mixed-type data with ordinal variables', Advances in Data Analysis and Classification. Available at: https://doi.org/10.1007/s11634-024-00595-5 (Accessed: 29 July 2026).

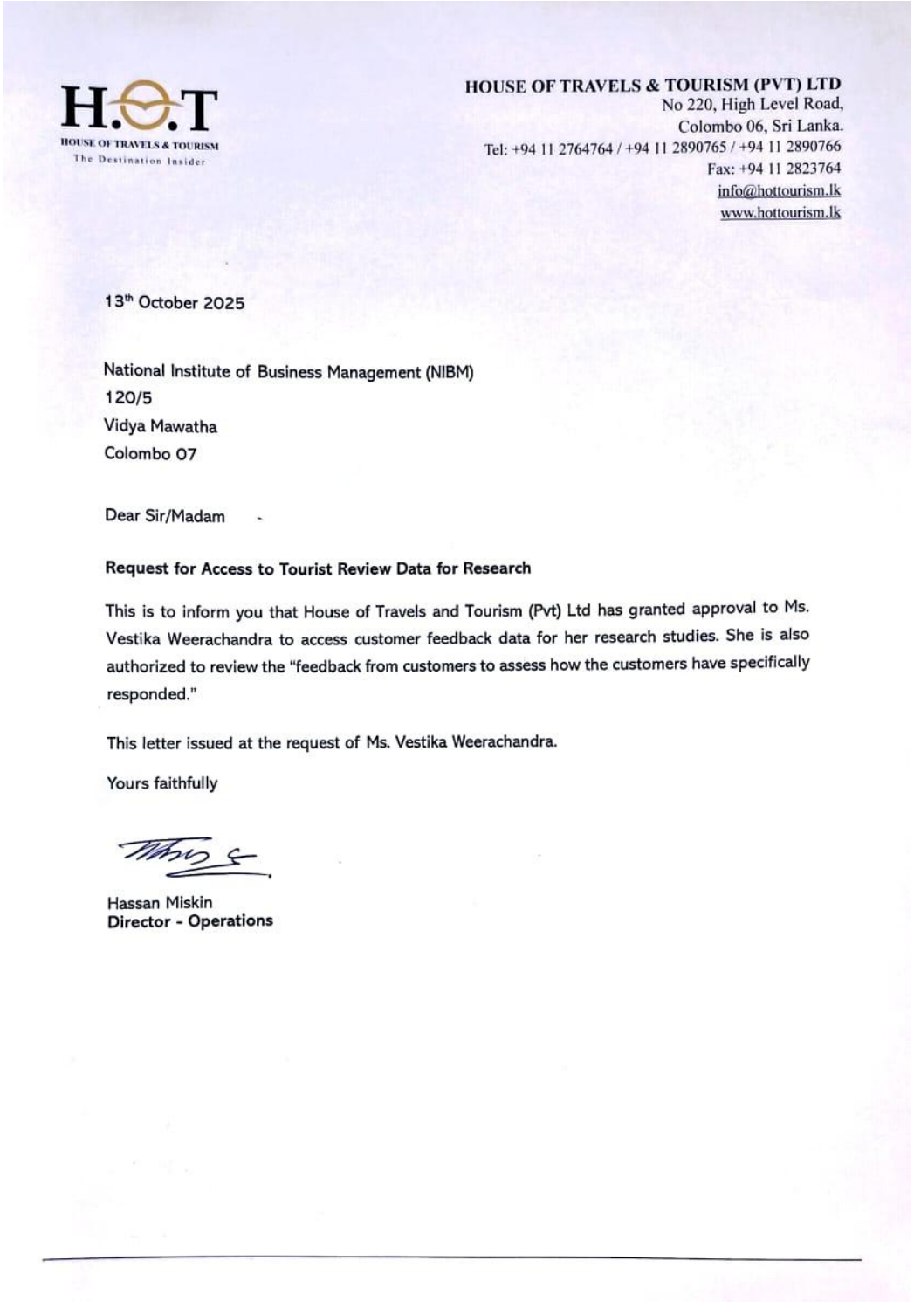
Van Leeuwen, R. and Koole, G. (2022) 'Data-driven market segmentation in hospitality using unsupervised machine learning', Machine Learning with Applications. Available at: https://doi.org/10.1016/j.mlwa.2022.100414 (Accessed: 25 July 2026).

Vargas-Calderon, V., Moros Ochoa, M.A., Castro Nieto, G. and Camargo, J. (2021) 'Machine learning for assessing quality of service in the hospitality sector based on customer reviews', Information Technology & Tourism. Available at: https://doi.org/10.1007/s40558-021-00207-4 (Accessed: 25 July 2026).

Walesiak, M. (1999) 'Distance Measure for Ordinal Data', Argumenta Oeconomica, 2(8), pp. 167-173.

Appendix

Appendix A: Approval letter from House of Travels & Tourism Pvt Ltd to use the data for research.



H.O.T
HOUSE OF TRAVELS & TOURISM
The Destination Insider

HOUSE OF TRAVELS & TOURISM (PVT) LTD
No 220, High Level Road,
Colombo 06, Sri Lanka.
Tel: +94 11 2764764 / +94 11 2890765 / +94 11 2890766
Fax: +94 11 2823764
info@hottourism.lk
www.hottourism.lk

13th October 2025

National Institute of Business Management (NIBM)
120/5
Vidya Mawatha
Colombo 07

Dear Sir/Madam

Request for Access to Tourist Review Data for Research

This is to inform you that House of Travels and Tourism (Pvt) Ltd has granted approval to Ms. Vestika Weerachandra to access customer feedback data for her research studies. She is also authorized to review the "feedback from customers to assess how the customers have specifically responded."

This letter issued at the request of Ms. Vestika Weerachandra.

Yours faithfully

Hassan Miskin
Director - Operations